Matrices & Vectors

ANSWER KEY

Section 1: Matrix Multiplication & Inverses

1 If A is a 3×2 matrix and B is a 2×4 matrix, what are the dimensions of the product matrix AB?

a. 3×4 ✓

b. 2×2

c. 4×3

d. Multiplication is undefined

Inner dimensions match ($2 = 2$). The resulting matrix has dimensions equal to outer indices: 3×4.

2 What is the inverse of the matrix $M = [[2, 5], [1, 3]]$?

a. $[[3, -5], [-1, 2]]$ ✓

b. $[[-3, 5], [1, -2]]$

c. $[[2, -5], [-1, 3]]$

d. $[[3, 1], [5, 2]]$

Determinant $\det(M) = 2(3) - 5(1) = 1$. The inverse is $\frac{1}{1}[[d, -b], [-c, a]] = [[3, -5], [-1, 2]]$.

3 Let A and B be invertible $n \times n$ matrices. Which expression represents $(AB)^{-1}$?

a. $B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}$

c. AB^{-1}

d. BA^{-1}

By standard matrix inverse rules, $(AB)^{-1} = B^{-1}A^{-1}$.

Section 2: Determinants & Cramer's Rule

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4 What is the determinant of the 3×3 matrix $[[1, 2, 3], [0, 4, 5], [0, 0, 6]]$?

a. 24 ✓

b. 30

c. 36

d. 0

The matrix is upper triangular. The determinant is the product of diagonal elements: $1 \times 4 \times 6 = 24$.

5 If the determinant of a 2×2 matrix A is 5, what is $\det(3A)$?

a. 15

b. 45 ✓

c. 75

d. 135

For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$. Here $n = 2$, so $\det(3A) = 3^2 \det(A) = 9 \times 5 = 45$.

6 For what value of x is the matrix $[[x, 3], [12, x]]$ singular?

a. ± 6 ✓

b. ± 4

c. ± 12

d. 0

Singular means determinant equals zero: $x^2 - 36 = 0 \Rightarrow x^2 = 36 \Rightarrow x = \pm 6$.

Section 3: Dot Product & Angle Between Vectors

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7 What is the angle θ between vectors $\mathbf{u} = \langle 1, 0 \rangle$ and $\mathbf{v} = \langle 1, 1 \rangle$?

a. 30°

b. 45° ✓

c. 60°

d. 90°

$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \frac{1(1)+0(1)}{(1)(\sqrt{2})} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$.

8 For what value of k are the vectors $\mathbf{a} = \langle 4, k \rangle$ and $\mathbf{b} = \langle 3, -6 \rangle$ orthogonal?

a. 1

b. 2 ✓

c. 3

d. 4

Orthogonal vectors have dot product equal to 0: $4(3) + k(-6) = 0 \Rightarrow 12 - 6k = 0 \Rightarrow 6k = 12 \Rightarrow k = 2$.

9 What is the scalar projection of vector $\mathbf{u} = \langle 4, 3 \rangle$ onto vector $\mathbf{v} = \langle 12, 5 \rangle$?

a. $\frac{63}{13}$ ✓

b. 5

c. $\frac{48}{13}$

d. $\frac{15}{13}$

Scalar projection $\text{comp}_{\mathbf{v}}\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|} = \frac{4(12)+3(5)}{\sqrt{12^2+5^2}} = \frac{48+15}{13} = \frac{63}{13}$.

Section 4: 3D Vectors & Cross Product

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10 What is the cross product $\mathbf{i} \times \mathbf{j}$ in three-dimensional space?

a. $\mathbf{k}$ ✓

b. $-\mathbf{k}$

c. $\mathbf{0}$

d. 1

By standard right-hand rule and unit vectors: $\mathbf{i} \times \mathbf{j} = \mathbf{k}$.

11 What is the magnitude of the vector $\mathbf{v} = \langle 2, -3, 6 \rangle$?

a. 7 ✓

b. 9

c. $\sqrt{49}$

d. 8

$\|\mathbf{v}\| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$

12 What is the area of the parallelogram determined by vectors $\mathbf{a} = \langle 3, 0, 0 \rangle$ and $\mathbf{b} = \langle 0, 4, 0 \rangle$?

a. 6

b. 12 ✓

c. 24

d. 7

Area equals the magnitude of the cross product: $\|\mathbf{a} \times \mathbf{b}\| = \|\langle 0, 0, 12 \rangle\| = 12.$

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